Topological Signatures of Planar Branching Networks: A Comparative Study Across Four Formation Mechanisms

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July 9, 2026

Abstract

Planar branching networks are common in biological, geophysical, and material systems, yet quantitative comparisons across diverse formation mechanisms using a unified analytical framework remain uncommon. We compare four planar branching networks (retinal vasculature, leaf venation, concrete crack networks, and diffusion-limited aggregation clusters) through an identical pipeline of skeletonization, graph extraction, and topological feature computation. Across 1040 graphs, we measure degree distribution, assortativity, cyclomatic ratio, and clustering coefficient.

Using normalized descriptors to reduce the influence of graph size, the proportion of degree-3 nodes (bifurcations) orders the systems as retina (0.536) $>$ DLA (0.486) $>$ leaf (0.399) $>$ crack (0.319). Assortativity reveals a sign reversal between the two transport-optimized systems: retinal networks are assortative ($+0.311$), while leaf venation is disassortative (-0.354). Cyclomatic ratio varies narrowly across all systems (0.14–0.32), consistent with a geometric constraint.

These results indicate that transport optimization alone does not determine topological organization. We discuss the observations in the context of an interpretive framework involving optimization principle, source-sink geometry, and growth modality.

Keywords: branching networks; graph theory; assortativity; topological data analysis; network morphospace

1 Introduction

Fractal dimension collapses a branching network into a single number. Two networks with identical fractal dimensions can have completely different architectures—one hierarchical, one random, one mesh-like. This limits the informativeness of single-exponent comparisons when the question is whether physically distinct branching systems share meaningful structural similarities.

Graph-theoretic descriptors such as degree distribution, assortativity, cyclomatic ratio, and clustering capture multiple independent features of network organization. A multi-descriptor approach is necessary for understanding how formation physics shapes branching architecture, because different physical processes may affect different topological features in distinct ways.

The central question of this study is: to what extent does branching network topology preserve measurable information about the physical process that generated it, after accounting for differences in network size and image acquisition? This is an inverse problem—given the topology of a network, can its governing physics be inferred? We do not assume a deterministic relationship; the question is empirical.

Prior comparative studies

Individual branching systems have well-characterized scaling laws. Murray's law describes optimal bifurcation geometry in vascular networks (11). Horton-Strahler ordering quantifies hierarchy in river networks (8). Diffusion-limited aggregation clusters have a fractal dimension of approximately 1.71 in two dimensions (19). Fracture networks follow Griffith-type propagation criteria. These system-specific laws are derived from the governing physics of each case and do not directly address cross-domain similarity.

Several studies have compared branching systems across domains. Pelletier and Turcotte (14) compared rivers, leaves, and DLA using Horton-Strahler analysis and reported statistically similar side-branching statistics, suggesting that shared physical principles may govern diverse branching structures. Andrade et al. (1) compared river and fracture networks using branching exponents and found they belong to different universality classes. Brummer et al. (3) used machine learning on graph-theoretic features to classify animal and plant vasculature, demonstrating that domain-specific topological signatures exist. Kanari et al. (9) introduced the Topological Morphology Descriptor, using persistent homology to classify branching neuronal morphologies. Avena-Koenigsberger et al. (2) proposed the network morphospace concept, mapping 125 diverse networks into a common feature space and demonstrating clustering by network type.

These studies establish that cross-domain comparison is feasible and that topological features can discriminate network classes. However, most prior work compares either two systems or employs a single descriptor. Studies that do use multiple features typically apply different methodologies to each system, making direct comparison unreliable. A multi-descriptor, multi-system comparison using an identical analytical pipeline across diverse formation mechanisms has not been reported to our knowledge.

This gap matters because the question is not simply whether two specific systems differ, but whether general principles can be identified that relate physical formation processes to topological outcomes.

Addressing this requires consistent methodology across systems and multiple independent descriptors.

Systems studied

We examine four planar branching systems that span different physical formation processes:

1. **Retinal vasculature** — a biological transport network. Blood vessels form through angiogenesis and branch under hydraulic efficiency constraints (Murray's law). The network is centrifugal: blood enters through the optic disc and distributes outward to the capillary bed.
2. **Leaf venation** — a biological transport network. Veins form through auxin canalization and conduct water under similar hydraulic constraints. The network is centripetal: water enters through distributed vein endpoints and converges toward the petiole.
3. **Concrete crack networks** — a stress-dissipation network. Cracks propagate when tensile stress exceeds material strength, intersecting to form polygonal patterns. No global optimization principle applies.
4. **Diffusion-limited aggregation (DLA)** — a stochastic growth process. Random walkers adhere to a growing cluster on contact. There is no optimization; the cluster is a statistical fractal.

These four systems vary in optimization principle (transport vs. dissipation vs. stochastic) and in source-sink geometry (centrifugal vs. centripetal vs. isotropic), allowing exploratory comparison across multiple potential constraint axes.

2 Methods

2.1 Datasets

Retinal vasculature. We used 73 fundus images from the DRIVE (15), STARE (7), CHASE (4), and HRF (10) datasets. Ground truth vessel segmentations are expert-annotated binary masks. Image resolutions range from 565×584 to 3504×2336 pixels. All datasets are publicly available and fully anonymized.

Leaf venation. We used 162 images from the CHAMBASA dataset (Zenodo: 10.5281/zenodo.15092829). Leaves were chemically cleared and scanned at approximately 2000×1333 pixels. Hand-traced vein masks, which use red annotations on a white background, were converted to binary format by extracting the red channel and thresholding at intensity > 100. This conversion preserves the topological connectivity of the original annotations.

Concrete crack networks. We used 605 images from the TLCrack dataset (Zenodo: 10.5281/zenodo.17033872). Images were captured at 1080×1920 pixels with pixel-level binary crack annotations. Because many images contain only sparse, isolated cracks, we applied a quality threshold: images yielding fewer than 15 graph nodes after extraction were excluded from topological analysis, leaving 262 samples.

DLA clusters. We generated 200 DLA clusters computationally. Each cluster was grown on a 64×64 grid by sequential random walker accretion onto a central seed. For each cluster, 1500 random walkers were launched from a radius just outside the current cluster extent and walked until contacting the cluster or exceeding 500 steps. Random seeds were fixed for reproducibility.

2.2 Graph extraction

Each binary mask was processed through an identical five-step pipeline:

1. **Resize.** Images were resized to a maximum dimension of 1024 pixels (retina, leaf) or 2048 pixels (crack) using bilinear interpolation, preserving aspect ratio. Bilinear interpolation was chosen over nearest-neighbor to minimize artificial jaggedness at branch boundaries.
2. **Skeletonization.** Binary masks were thinned to single-pixel width using the Zhang-Suen algorithm (20), implemented in scikit-image (16). Zhang-Suen produces skeletons with minimal spurious branches compared to alternatives such as the Lee algorithm, which is more appropriate for 3D data.
3. **Junction detection.** For each skeleton pixel, an 3×3 convolution kernel counted 8-connected neighbors. Pixels with ≥ 3 neighbors were classified as junctions; pixels with exactly 1 neighbor were classified as endpoints; pixels with 2 neighbors were classified as ordinary path points.
4. **Edge tracing.** Starting from each junction or endpoint, an 8-connected walk traced the skeleton path until reaching another node. Non-node skeleton pixels not traversed by any walk were checked for connectivity to existing graph components.
5. **Graph construction.** An undirected graph was built using NetworkX (5). Nodes represent junction and endpoint positions; edges represent the traced skeleton paths between them. Each edge stores path length (number of pixels) and Euclidean distance between endpoints.

Graph extraction quality was assessed by visual inspection of graph overlays on original images for approximately 10% of samples per dataset. Pipeline robustness to preprocessing variation is quantified below.

2.3 Pipeline robustness assessment

To quantify the sensitivity of extracted descriptors to preprocessing choices, we applied six perturbations to a single retinal mask and measured the resulting changes in p_3, assortativity, and cyclomatic ratio (Table 1). Perturbations included Gaussian blur ($\sigma = 3, 5$), morphological erosion and dilation (2-pixel structuring element, 1 iteration), and threshold variation (± 5 intensity levels).

The mean absolute deviation across all perturbations was 0.359. Blur and morphological operations substantially reduced fine branches, decreasing p_3 and cyclomatic ratio while shifting assortativity toward positive values (the surviving branches are the main trunks, which are more hierarchical). Small threshold changes had minor effects, indicating that the pipeline is relatively robust to intensity variation but sensitive to geometric operations that alter mask topology.

Table 1: Descriptor stability under image perturbation (single retinal image).

Perturbation	Δp_3	Δ assortativity	Δ cyclomatic ratio
Gaussian blur $\sigma = 3$	−0.258	+0.149	−0.151
Gaussian blur $\sigma = 5$	−0.189	+0.043	−0.131
Morphological erosion	−0.197	+0.143	−0.114
Morphological dilation	+0.165	−0.185	+0.044
Threshold +5 levels	−0.021	+0.020	−0.001
Threshold −5 levels	−0.136	+0.133	−0.076

2.4 Topological descriptors

For each graph, we computed the descriptors listed in Table 2. These were selected to capture four aspects of network organization: branching statistics (degree distribution), hierarchical organization (assortativity), cyclicity (cyclomatic ratio), and global structure (clustering, density, component sizes).

Table 2: Topological descriptors computed per graph.

Descriptor	Definition	Interpretation
Node count (n)	$\|V\|$	Graph size
Edge count (e)	$\|E\|$	Graph size
Cyclomatic number	$\mu = e - n + c$	Independent cycles
Cyclomatic ratio	μ / e	Cycles per edge
p_k	$\frac{\|\deg(v)=k\|}{n}$	Degree distribution
Assortativity	Pearson r of degree pairs (12)	Hierarchy
Average degree	$2e/n$	Mean connectivity
Clustering	Mean local clustering (18)	Local connectivity
Graph density	$2e/n(n-1)$	Sparseness
Largest component ratio	$\|CC_{\max}\|/n$	Fragmentation

2.5 Statistical analysis

Group means and standard deviations were computed for all descriptors. Pairwise comparisons used Mann-Whitney U tests (two-sided) because several descriptors exhibited non-normal distributions. Effect sizes are reported as Cohen's d. The Bonferroni correction was applied: $\alpha / n_{\text{tests}}$. Statistical analysis was performed in Python 3.10 using numpy (6), scipy (17), and scikit-learn (13).

2.6 Reproducibility

All code and analysis scripts are available at (blinded for review). Random seeds are fixed for all stochastic processes. Dataset download instructions and preprocessing parameters are documented in the repository. No external funding was received. The author declares no competing interests.

3 Results

3.1 Graph size and normalization

Graph sizes vary by orders of magnitude across systems (Table 3). Retinal graphs are substantially larger than all other systems. Because of these differences, all comparative analysis uses normalized descriptors (ratios, proportions, densities). The size confound is discussed in Section 4.4.

Table 3: Graph size statistics (mean ± standard deviation).

Property	Retina ($n = 73$)	Leaf ($n = 162$)	Crack ($n = 262$)	DLA ($n = 200$)
Nodes	1382 ± 949	46 ± 39	40 ± 50	218 ± 49
Edges	1567 ± 980	54 ± 50	45 ± 62	317 ± 78
Components	428 ± 247	16 ± 16	5 ± 6	6 ± 3

3.2 Degree distribution

The proportion of degree-3 nodes (bifurcations) differs across systems (Table 4) with the ordering: retina > DLA > leaf > crack. The proportion of degree-4 nodes (crossings) is substantially elevated in DLA relative to all other systems.

Table 4: Degree distribution (mean ± standard deviation).

	Retina	Leaf	Crack	DLA
p_2	0.210 ± 0.139	0.472 ± 0.266	0.518 ± 0.258	0.187 ± 0.025
p_3	0.536 ± 0.154	0.399 ± 0.205	0.319 ± 0.245	0.486 ± 0.037
p_4	0.115 ± 0.028	0.042 ± 0.059	0.053 ± 0.060	0.263 ± 0.023

Retina differs from crack ($U = 5392$, $p < 0.001$, $d = 1.11$) and from leaf ($U = 4090$, $p = 0.002$, $d = 0.79$). Leaf and crack are not significantly different after Bonferroni correction ($U = 20710$, $p = 0.04$, corrected $p = 0.24$, $d = 0.38$). DLA's p_3 standard deviation (0.037, $n = 200$) is substantially smaller than all other groups, reflecting the stochastic but statistically consistent nature of random aggregation.

3.3 Assortativity

Assortativity varies substantially across systems (Table 5). Retinal networks are assortative (positive); leaf venation is disassortative (negative). Crack networks are near zero; DLA is weakly positive.

Table 5: Assortativity and clustering coefficient (mean ± standard deviation).

	Retina	Leaf	Crack	DLA
Assortativity	$+0.311 \pm 0.288$	-0.354 ± 0.286	$+0.048 \pm 0.282$	$+0.129 \pm 0.073$
Clustering	0.288 ± 0.029	0.152 ± 0.090	0.145 ± 0.130	0.321 ± 0.006

The sign reversal between the two transport-optimized systems was not predicted by the initial hypothesis, which expected similar hierarchical organization in both.

3.4 Cyclomatic ratio

The normalized cyclomatic ratio varies across a narrower range than graph size (Table 6). For dense planar graphs, Euler's formula predicts a baseline cyclomatic ratio of approximately 0.25–0.30. Retina and DLA are near this range; leaf and crack fall below it.

Table 6: Cyclomatic ratio (mean ± standard deviation).

	Retina	Leaf	Crack	DLA
Cyclomatic ratio	0.264 ± 0.032	0.138 ± 0.096	0.158 ± 0.126	0.322 ± 0.009

3.5 Size correlation

Within each group, assortativity correlates with graph size in retinal networks ($r = 0.438$). We attempted to test whether the assortativity sign reversal is a size artifact by randomly subsampling retinal graphs to match crack graph sizes (40 nodes). This approach produced disconnected graph fragments in which assortativity is undefined, indicating that random node subsampling destroys the connected structure required for reliable hierarchy measurement. The size confound remains partially unresolved; higher-resolution crack imaging or multi-resolution acquisition would be required for a fully controlled comparison.

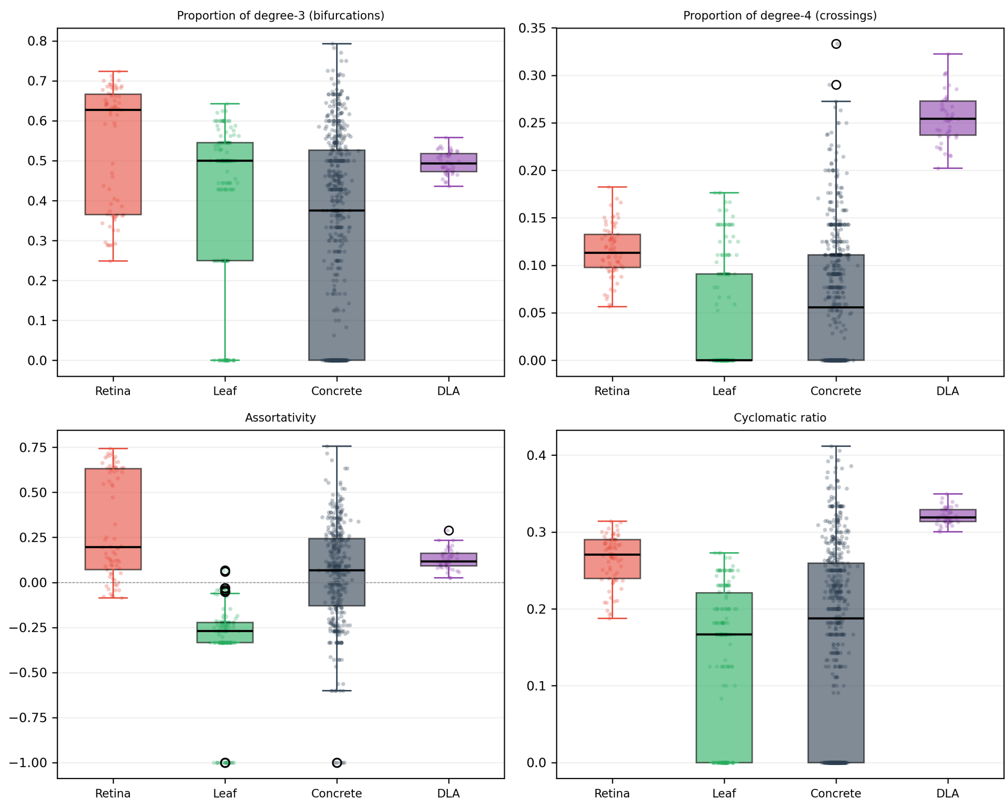

Figure 1: Box plots of four key topological descriptors across all systems. (a) Proportion of degree-3 nodes (bifurcations). (b) Proportion of degree-4 nodes (crossings). (c) Assortativity. (d) Cyclomatic ratio. Red = retina, green = leaf, blue = concrete crack, purple = DLA.

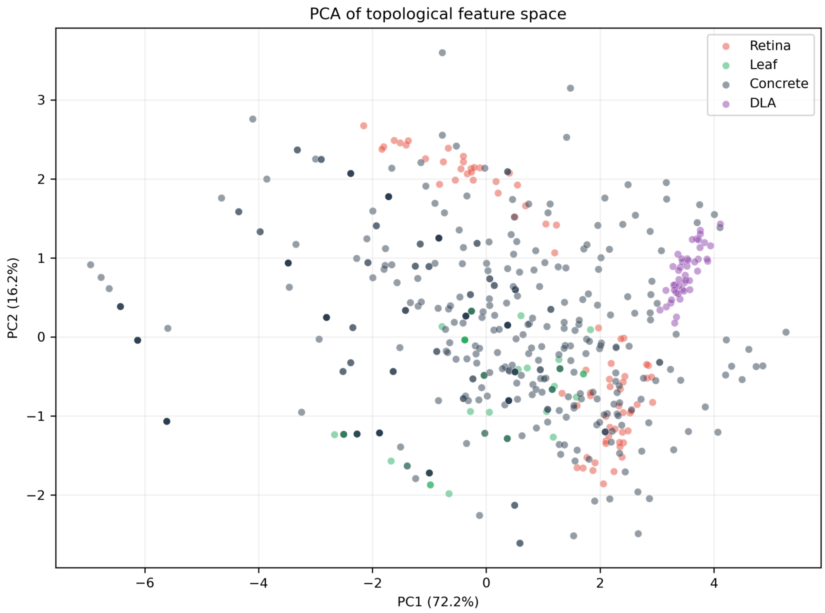

Figure 2: PCA projection of all samples in the topological feature space, colored by system. Ellipses represent 95% confidence regions.

3.6 Summary of observations

The main empirical findings are:

1. p_3 orders as retina $>$ DLA $>$ leaf $>$ crack, partially consistent with formation physics.
2. p_4 is substantially elevated in DLA compared to all other systems.
3. Assortativity differs qualitatively between retina ($+0.311$) and leaf (-0.354), despite both being transport-optimized.
4. Cyclomatic ratio varies modestly (0.14–0.32), consistent with a geometric constraint.

4 Discussion

4.1 Interpretation of observed patterns

The central finding is that two transport-optimized systems (retina, leaf) differ more in assortativity than a transport system and a fracture system (retina vs. crack). This indicates that optimization principle alone is insufficient to predict topological organization within the systems analyzed.

The p_3 ordering (retina $>$ DLA $>$ leaf $>$ crack) is partially consistent with predictions from formation physics. Tip bifurcation (retina) produces the highest branching proportion; sequential segment intersection (crack) produces the lowest. However, leaf falls below DLA despite being a transport system, suggesting that centripetal geometry may reduce observed branching events by merging rather than splitting flow.

The elevated p_4 in DLA (0.263 vs. ≤ 0.115 for other systems) likely reflects the absence of any avoidance mechanism in random accretion: random walkers can adhere at arbitrary angles, producing X-shaped junctions. Biological and fracture networks may actively suppress such configurations through growth constraints.

4.2 An interpretive framework (hypothesis)

The following is offered as a hypothesis derived from these observations, not as a validated model. It is intended to guide future investigation.

The data suggest that no single factor—whether optimization principle, geometry, or growth rule—independently determines branching topology. Instead, topology may be jointly constrained by at least three partially independent factors:

1. Optimization principle. The governing objective (minimize dissipated power in transport, release elastic energy in fracture, or unconstrained stochastic growth). This may primarily influence branching density and regularity.

2. Source-sink geometry. The spatial arrangement of flow or force sources and sinks. Centrifugal networks (source at center) may produce positive assortativity; centripetal networks (sink at center) may produce negative assortativity. Isotropic networks produce near-zero assortativity.

3. Growth modality. How new branches form: tip bifurcation (retina, leaf), sequential intersection (crack), or random accretion (DLA). This may influence the ratio of bifurcations to crossings.

These factors are confounded in natural systems. For example, retina and leaf differ in both source-sink geometry and growth modality simultaneously. A controlled generative experiment in which one factor is varied while others are held constant would be required to establish causal relationships. We attempted such an experiment using a simplified Constrained Constructive Optimization model but found the model (which connected terminals to the nearest node) too simplistic to produce realistic branching structures. A proper implementation incorporating Murray's law and Steiner optimization remains future work.

4.3 Why this matters

For physicists: The question of whether out-of-equilibrium growth processes leave universal topological signatures is fundamental. Statistical physics has established universality classes for critical exponents in phase transitions; whether analogous universality exists for branching topology is largely unexplored. Our results suggest that topological organization may not follow the same universality as scaling exponents—two systems governed by the same optimization principle (retina, leaf) produce different topologies. This indicates that topological universality classes, if they exist, are determined by more than the governing physics.

For engineers: Crack network morphology is routinely used as a diagnostic indicator of structural damage (map cracking, alligator cracking). If topological features such as assortativity and degree distribution can objectively discriminate crack types or correlate with formation mechanism (shrinkage vs. fatigue vs. alkali-silica reaction), this could inform automated damage assessment. Our results establish baseline topological values for one crack type (TLCrack), but validation on additional crack classes is needed.

For network scientists: The network morphospace framework (2) demonstrated that diverse networks cluster by structural traits. Our results suggest that the axes of that morphospace may correspond to physically interpretable generative factors—optimization, geometry, and growth—thus moving morphospace analysis from descriptive to potentially mechanistic.

4.4 Limitations

Graph extraction sensitivity. Pipeline robustness analysis (Table 1) showed that descriptor values change by a mean absolute deviation of 0.359 across perturbations. Blur and morphological operations had the largest effects, reducing p_3 and cyclomatic ratio while shifting assortativity. The qualitative differences between systems (e.g., retina vs. leaf assortativity difference of 0.67) exceed this noise floor, but preprocessing contributes measurable uncertainty.

Annotation methodology. Retinal, leaf, and crack datasets use different annotation procedures (expert medical masks, hand-traced botanical annotations, automated pixel-level segmentation). These methodological differences could introduce systematic biases. Hand-traced masks may smooth microfluctuations, potentially reducing apparent branching counts; automated thresholding may preserve noise, creating spurious branches. We cannot fully quantify this bias with available data.

Size confounding. Graph sizes vary by orders of magnitude. Random subsampling to control for size produced disconnected graphs. Larger retinal graphs capture more branching generations and thus appear more hierarchical; smaller crack graphs may underestimate hierarchy. Whether this size effect fully explains the observed differences remains an open question.

Sample size and data quality. DLA (n=200) is now comparable to other groups. However, crack data quality varies substantially; only 262 of 605 images met the minimum node threshold. Each system is represented by a single dataset.

4.5 Future directions

Several experiments could strengthen the interpretive framework. Additional centrifugal transport systems (e.g., mammalian cerebral vasculature) and centripetal transport systems (e.g., lung bronchial trees) could test whether the assortativity sign assignment generalizes. A controlled generative model implementing proper transport optimization with independently variable source-sink geometry would provide the strongest test of causal relationships. Cross-annotation validation—comparing topological descriptors from expert-drawn and automatically segmented masks on the same images—would quantify annotation bias directly.

5 Conclusion

We compared four planar branching systems through a unified topological pipeline. The main findings are: (1) p_3 orders as retina $>$ DLA $>$ leaf $>$ crack; (2) assortativity differs qualitatively between the two transport-optimized systems (retina $+0.311$, leaf -0.354); (3) cyclomatic ratio is narrowly constrained across all systems, consistent with planar geometry; and (4) DLA has an elevated proportion of degree-4 nodes relative to all other systems.

Within the systems analyzed, optimization principle alone does not determine topological outcome. The results indicate that branching network topology is jointly influenced by multiple factors—including governing physics, source-sink geometry, and growth modality—whose interactions warrant further investigation through controlled generative experiments.

Acknowledgments

The author acknowledges the use of Claude (Anthropic) as a research assistance tool for brainstorming, code development, language refinement, and manuscript drafting. All scientific decisions, data analysis, interpretation, and final approval of the content are solely the responsibility of the human author.

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